AWS 기본

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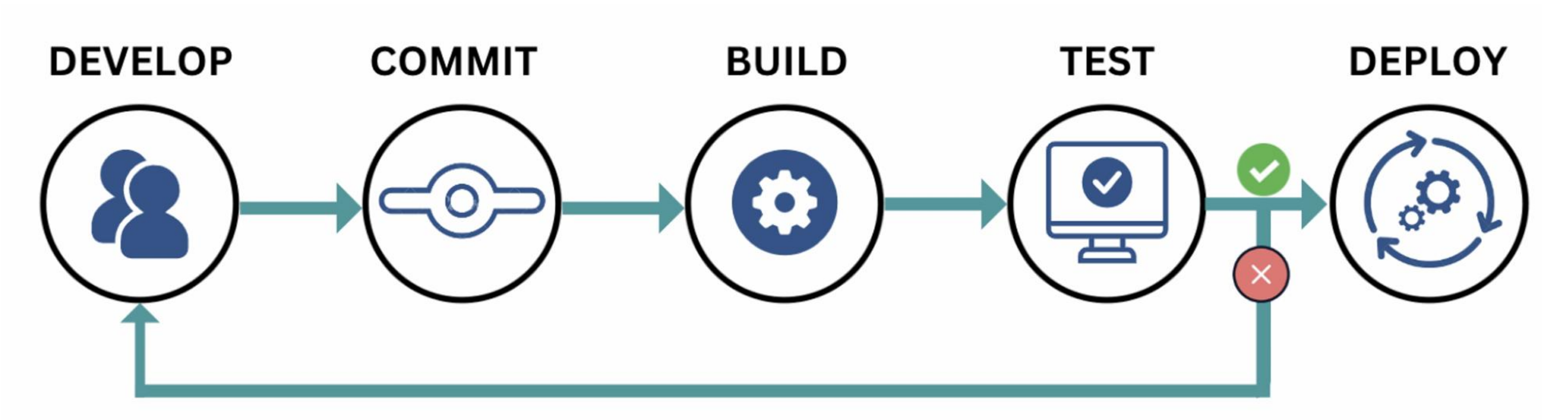
16. Github Actions

16.1 CI/CD

개념

- Continuous Integration (지속적 통합)
- Continuous Deployment (지속적 배포)



16.2 Github Actions

1. 개요

- https://docs.github.com/en/actions
- Github에서 제공하는 워크플로우(workflow) 자동화 서비스.
- 여러 Repository 관련 프로세스와 작업을 자동화 지원.
- 빌드, 테스트 및 배포 파이프라인을 자동화할 수 있는 CI/CD 플랫폼.
- 저장소에 대한 PR(Pull Request) 요청을 빌드 및 테스트하거나 병합된 PR 요청에 배포하는 워크플로우를 만들 수 있음.

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2. Github Actions 구성요소

- workflow
  - 하나 이상의 작업을 실행할 구성 가능한 자동화된 프로세스임
  - 순차/병렬 모두 가능.
  - 하나 이상의 job 포함.
  - 저장소의 이벤트 또는 수동 및 스케쥴러에 의해 트리거됨.
  - 미리 구성된 워크플로우 템플릿을 제공함.
- event
  - 워크플로우 실행을 트리거하는 저장소의 특정 활동임.
  - ex. PR 요청, 커밋 push, 스켜쥴러(일정)
- job
  - 동일한 실행기(runner)에서 실행되는 워크플로우의 단계(step) 집합임.
  - 단계(step)는 실행되는 쉘스크립트 또는 실행되는 작업 의미.
  - 단계(step)는 순서대로 실행되고 서로 종속적임.
  - 동일한 실행기(runner)에서 실행되기 때문에 단계간에 데이터 공유가 가능함.
  - 기본적으로 작업(job)은 병렬로 실행되지만 needs 속성을 이용해서 순차로 실행 강제 가능.

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- action
  - 복잡하고 자주 반복하는 task를 수행하는 사용자 지정 어플리케이션임.
  - action을 사용하여 워크플로우 파일에 작성하는 반복 코드양을 감소시킬 수 있음.

- runner
  - 이벤트가 트리거될 때 워크플로우를 실행하는 서버.
  - 한 번에 단 하나의 job만 실행 가능.
  - Github는 워크플로우를 실행할 Linux, Window, MacOS를 제공.
  - 각 워크플로우는 새로 프로비저닝된 새 가상머신에서 실행됨.

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3. 기본 구조

https://github.com/actions/starter-workflows/blob/main/ci/blank.yml

```
# This is a basic workflow to help you get started with Actions

name: CI

# Controls when the workflow will run
on:
  # Triggers the workflow on push or pull request events but only for the $default-branch branch
  push:
    branches: [ $default-branch ]
  pull_request:
    branches: [ $default-branch ]

  # Allows you to run this workflow manually from the Actions tab
  workflow_dispatch:

# A workflow run is made up of one or more jobs that can run sequentially or in parallel
jobs:
  # This workflow contains a single job called "build"
  build:
    # The type of runner that the job will run on
    runs-on: ubuntu-latest
```

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```
# Steps represent a sequence of tasks that will be executed as part of the job
  steps:
    # Checks-out your repository under $GITHUB_WORKSPACE, so your job can access it
    - uses: actions/checkout@v4

    # Runs a single command using the runners shell
    - name: Run a one-line script
      run: echo Hello, world!

    # Runs a set of commands using the runners shell
    - name: Run a multi-line script
      run: |
        echo Add other actions to build,
        echo test, and deploy your project.
```

16.3 Github Actions 실습

실습1

```yaml
name: First workflow
on: workflow_dispatch
jobs:
  first-job:
    runs-on: ubuntu-latest
    steps:
      - name: Print greeting
        run: echo "Hello World"
      - name: Print Good Bye
        run: |
         echo "완료1"
         echo "완료2"
```

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실습2

https://github.com/marketplace/actions/checkout

```
name: First workflow2
on: push
jobs:
  first-job2:
    runs-on: ubuntu-latest
    steps:
      - name: github checkout
        # https://github.com/marketplace/actions/checkout 참조
        uses: actions/checkout@v4
      - name: Run a one-line script
        run:  echo " The ${{ github.repository }} repository has been cloned to the
runner."
```

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실습3

https://github.com/marketplace/actions/setup-node-js-environment

```yaml
name: First workflow3
on: push
jobs:
  first-job3:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v4

      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 18

      - name: install dependencies
        run: npm install axios

      - name: Run a one-line script
        run: echo " The ${{ github.repository }} repository has been cloned to the runner."
```

16.3 Github Actions 실습

실습4

https://docs.github.com/ko/actions/writing-workflows/choosing-what-your-workflow-does/accessing-contextual-information-about-workflow-runs

```yaml
name: Context testing
on: push

jobs:
  dump_contexts_to_log:
    runs-on: ubuntu-latest
    steps:
      - name: Dump GitHub context
        env:
          GITHUB_CONTEXT: ${{ toJson(github) }}
        run: echo "$GITHUB_CONTEXT"
      - name: Dump job context
        env:
          JOB_CONTEXT: ${{ toJson(job) }}
        run: echo "$JOB_CONTEXT"
```

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```yaml
      - name: Dump steps context
        env:
          STEPS_CONTEXT: ${{ toJson(steps) }}
        run: echo "$STEPS_CONTEXT"
      - name: Dump runner context
        env:
          RUNNER_CONTEXT: ${{ toJson(runner) }}
        run: echo "$RUNNER_CONTEXT"
      - name: Dump strategy context
        env:
          STRATEGY_CONTEXT: ${{ toJson(strategy) }}
        run: echo "$STRATEGY_CONTEXT"
      - name: Dump matrix context
        env:
          MATRIX_CONTEXT: ${{ toJson(matrix) }}
        run: echo "$MATRIX_CONTEXT"
```

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실습5

https://docs.github.com/en/actions/writing-workflows/about-workflows#creating-dependent-jobs

```yaml
name: Multi Job workflow
on: push
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v4
      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 20
      - name: install dependencies
        run: npm install axios
      - name: Run Test
        run: echo Run Test
```

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```yaml
deploy:
  runs-on: ubuntu-latest
  steps:
    - name: Checkout
      uses: actions/checkout@v4
    - name: Set up Node.js
      uses: actions/setup-node@v4
      with:
        node-version: 20
    - name: install dependencies
      run: npm install axios
    - name: Run Deploy
      run: echo Run Deploy
```

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실습6

https://docs.github.com/en/actions/writing-workflows/about-workflows#creating-dependent-jobs

```yaml
name: Multi Job workflow-needs
on: push
jobs:
  test:
    needs: deploy
    runs-on: ubuntu-latest
    steps:
      - name: Checkout
        uses: actions/checkout@v4
      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: 20
      - name: install dependencies
        run: npm install axios
      - name: Run Test
        run: echo Run Test
```

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```yaml
deploy:
  runs-on: ubuntu-latest
  steps:
    - name: Checkout
      uses: actions/checkout@v4
    - name: Set up Node.js
      uses: actions/setup-node@v4
      with:
        node-version: 20
    - name: install dependencies
      run: npm install axios
    - name: Run Deploy
      run: echo Run Deploy
```

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실습7

https://docs.github.com/ko/actions/writing-workflows/choosing-what-your-workflow-does/running-variations-of-jobs-in-a-workflow

```yaml
name: Matrix workflow
on: push
jobs:
  test:
    strategy:
      matrix:
         os: [ubuntu-latest , windows-latest]
         version: [18,30,20]
    continue-on-error: true
    runs-on: ${{matrix.os}}
    steps:
      - name: Checkout
        uses: actions/checkout@v4
      - name: Set up Node.js
        uses: actions/setup-node@v4
        with:
          node-version: ${{matrix.version}}
      - name: install dependencies
        run: npm install axios
      - name: Run Test
```

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실습8

https://docs.github.com/en/actions/writing-workflows/choosing-what-your-workflow-does/running-variations-of-jobs-in-a-workflow#example-expanding-configurations

```yaml
name: Matrix workflow
on: push
jobs:
  test:

    strategy:
      matrix:
        os: [ubuntu-latest, windows-latest]
        version: [18, 20]
        include:
          - os: ubuntu-latest
            version: 19
        exclude:
          - os: ubuntu-latest
            version: 18
    continue-on-error: true
```

```yaml
runs-on: ${{ matrix.os }}
steps:
  - name: Checkout
    uses: actions/checkout@v4
  - name: Set up Node.js
    uses: actions/setup-node@v4
    with:
      node-version: ${{ matrix.version }}
  - name: Install dependencies
    run: npm install axios
  - name: Run Test
    run: echo Run Test
```

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실습9

https://docs.github.com/ko/actions/writing-workflows/choosing-when-your-workflow-runs/using-conditions-to-control-job-execution

```yaml
name: if action workflow
on: push
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Run Test
        if: runner.os == 'Linux'
        run: |
          ls ${{ github.workspace }}
          echo Run Test
```

```yaml
build:
  if: github.repository == 'inky4832/github-action-test'
  needs: test
  runs-on: ubuntu-latest
  steps:
    - name: Run Build
      run: |
        ls ${{ github.workspace }}
        echo Run Build
deploy:
  needs: build
  runs-on: ubuntu-latest
  steps:
    - name: Run Deploy1
      id: run_deploy1
      run: echo Run Deploy1
    - name: Run Deploy2
      if: steps.run_deploy1.outcome == 'success'
      run: echo Run Deploy2
```

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실습10

https://docs.github.com/ko/actions/writing-workflows/choosing-when-your-workflow-runs/events-that-trigger-workflows

https://docs.github.com/ko/actions/writing-workflows/choosing-when-your-workflow-runs/triggering-a-workflow

```
name: advanced event action workflow

on: [push, pull_request, issues, fork, workflow_dispatch]

jobs:

  test:
    runs-on: ubuntu-latest

    steps:
      - name: Run Test
        run: |
          echo github.event_name: "${{ github.event_name }}"
          echo Run Test
```

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실습11

https://docs.github.com/en/actions/writing-workflows/workflow-syntax-for-github-actions#using-activity-types

```
name: advanced event activity type action workflow
on:
    push:
    issues:
        types:
        - opened
        - edited
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Run Test
        run: |
          echo github.event_name: "${{ github.event_name }}"
          echo Run Test
```

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실습12

ttps://docs.github.com/ko/actions/writing-workflows/workflow-syntax-for-github-actions#using-filters

```yaml
name: advanced event filters action workflow
on:
    push:
      branches:
        - 'hotfix'
jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - name: Run Test
        run: |
          echo github.event_name: "${{ github.event_name }}"
          echo Run Test
```

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실습13

https://docs.github.com/ko/actions/writing-workflows/choosing-when-your-workflow-runs/events-that-trigger-workflows#running-your-workflow-only-when-a-push-affects-specific-files

```yaml
name: advanced event filters action workflow

on:
    push:
      paths:
          - '**.js'
jobs:

  test:
    runs-on: ubuntu-latest

    steps:
      - name: Run Test
        run: |
          echo github.event_name: "${{ github.event_name }}"
          echo Run Test
```

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실습14

https://docs.github.com/en/actions/writing-workflows/choosing-what-your-workflow-does/store-information-in-variables#default-environment-variables

```
name: env action workflow
on: push
# workflow 레벨
env:
  DAY_OF_WEEK: Monday
jobs:
  greeting_job:
    runs-on: ubuntu-latest
    # job 레벨
    env:
      Greeting: Hello
    steps:
        - name: Run greeting_job1
          # steps 레벨
          env:
              First_Name: Mona
          if: ${{ env.DAY_OF_WEEK == 'Monday' }}
          run: |
              echo  greeting_job1 stpe LEVEL env : "$First_Name" , "${{env.First_Name}}"
              echo  greeting_job1 job LEVEL env : "$Greeting" , "${{env.Greeting}}"
              echo  greeting_job1 workflow LEVEL env : "$DAY_OF_WEEK" , "${{env.DAY_OF_WEEK}}"
```

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실습15

https://docs.github.com/en/actions/security-for-github-actions/security-guides/using-secrets-in-github-actions

```
name: secrets env action workflow
on: push
# workflow 레벨
env:
  USER_NAME: ${{ secrets.USER_NAME }}
  USER_PW:   ${{ secrets.USER_PW }}
jobs:
  secrets_env_variable:
    runs-on: ubuntu-latest
    steps:
        - name: Run secrets env
          run: |
              echo  USER_NAME = "${{env.USER_NAME}}"
              echo  USER_PW   = "${{env.USER_PW}}"
```

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실습16

https://docs.github.com/en/actions/security-for-github-actions/security-guides/using-secrets-in-github-actions

```
name: Environments secrets env action workflow
on: push
jobs:
  test:
    runs-on: ubuntu-latest
    environment: local
    env:
        DB_NAME1:   ${{ secrets.DB_NAME }}
        DB_PORT1:   ${{ vars.DB_PORT }}
    steps:
        - name: Run Environments secrets env
          run: |
              echo local DB_NAME =   "${{env.DB_NAME1}}"
              echo local DB_PORT =   "${{env.DB_PORT1}}"
```

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```yaml
prod:
  runs-on: ubuntu-latest
  environment: prod
  env:
      DB_NAME2:   ${{ secrets.DB_NAME }}
      DB_PORT2:   ${{ vars.DB_PORT }}
  steps:
      - name: Run Environments secrets env
        run: |
            echo prod DB_NAME =   "${{env.DB_NAME2}}"
            echo prod DB_PORT =   "${{env.DB_PORT2}}"
```

수고하셨습니다.